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SUPPLEMENTARY METHODS

Drug nomenclature

The radiopharmaceutical name for ^{177}Lu-PSMA-617 is [^{177}Lu]Lu-PSMA-617 and the international nonproprietary name is lutetium (^{177}Lu) vipivotide tetraxetan. The radiopharmaceutical name for ^{68}Ga-PSMA-11 is [^{68}Ga]Ga-PSMA-11 and the international non-proprietary name is gallium (^{68}Ga) gozetotide.

Study drug administration

Patients randomised to the ^{177}Lu-PSMA-617 group received ^{177}Lu-PSMA-617 7·4 GBq (200 mCi) (±10%) once every 6 weeks for six cycles. Physical examinations, toxicity assessments, and haematology and blood biochemistry results were evaluated before each cycle. ^{177}Lu-PSMA-617 was administered slowly via the intravenous route, preceded and followed by saline flush. Vital signs were monitored 15 minutes before, and 30 and 60 minutes after, administration. Patients were encouraged to maintain good fluid intake. Preparation and administration of ^{177}Lu-PSMA-617 was in accordance with pharmaceutical quality requirements and radiation safety regulations. Patients randomised to the ARPI change group received abiraterone or enzalutamide orally on a continuous basis, as per product labelling and guidelines.

Permitted and prohibited concomitant treatments

Androgen deprivation therapy was continued and concomitant supportive care was permitted, including bone health agents. Prohibited concomitant treatments were investigational drugs, biological products, chemotherapy, immunotherapy, PARP inhibitors, systemic radioisotopes, and hemi-body radiotherapy. Concurrent treatment with an ARPI and ^{177}Lu-PSMA-617 was prohibited in both randomised and crossover groups.

Treatment after disease progression

Upon confirmed radiographic progression by BICR, participants randomised to the ^{177}Lu-PSMA-617 group could either stop receiving ^{177}Lu-PSMA-617 and receive another therapy according to local regulations, or continue receiving ^{177}Lu-PSMA-617 for up to six cycles if the patient was determined to still be benefiting from treatment, as per the discretion of the treating physician. Participants randomised to the ARPI change group who crossed over to receive ^{177}Lu-PSMA-617 upon confirmed radiographic progression received ^{177}Lu-PSMA-617 using the same dosage and schedule as for participants randomised to the ^{177}Lu-PSMA-617 group.

Crossover eligibility criteria

Participants randomised to the ARPI change group were allowed to crossover to receive ^{177}Lu-PSMA-617 at the discretion of the treating physician, within 28 days of confirmed radiographic progression by BICR as per PCWG3-modified RECIST v1·1. Participants were eligible for crossover if they had received no intervening antineoplastic therapy after their randomised treatment, had controlled unresolved toxicity from their previous therapy of grade 2 or less, had an ECOG performance status of 0–1, and had adequate organ function.

Dose modifications

Dose escalations and modifications in the ARPI change group were managed as per the local labelling and/or local practice guidelines for abiraterone and enzalutamide. Dose modifications in the ^{177}Lu-PSMA-617 group were carried out at the discretion of the investigator according to dose management guidance. One reduction in administered ^{177}Lu-PSMA-617 activity was permitted when an adverse event was deemed related to ^{177}Lu-PSMA-617; reduced doses were not re-escalated. Delays of more than 4 weeks to ^{177}Lu-PSMA-617 therapy because of an adverse event or unacceptable toxicity led to ^{177}Lu-PSMA-617 discontinuation. Details of toxicity management and dose modifications are shown in Table 6·2 of the trial protocol (available at thelancet.com).

Censoring and missing data

Participants who were alive and without radiographic progression at the time of data cut-off or who were lost to follow-up at the time of analysis were censored for radiographic progression-free survival at the time of their last imaging assessment. Patients with two or more consecutive missed scans immediately before disease progression or death were censored at the date of their last evaluable scan before the two missed scans.

Prespecified subgroup analyses

Subgroup analysis of radiographic progression-free survival by prespecified demographic and baseline characteristic parameters was planned. Prespecified subgroups are detailed in the statistical analysis plan. No inferential statistical tests were performed in subgroups.

RECIST v1·1 responses

RECIST v1·1-based endpoints were analysed as per PCWG3 guidelines in patients with measurable or evaluable disease at baseline. Measurable disease at baseline was defined as the presence of at least one target lesion. Evaluable disease was defined as the presence of at least one target or non-target lesion. Patients with target lesions were classified as complete response (CR), partial response (PR), stable disease (SD), progressive disease (PD), or non-evaluable (NE). Patients with non-target lesions only were additionally classified as non-CR/non-PD. Objective response rate was the proportion of patients who had a best overall response of CR or PR. Disease control rate was the proportion of patients who had a best overall response of CR, PR, or SD (or non-CR/non-PD in patients with non-target lesions only).

Statistical analyses of patient-reported outcomes

Health-related quality of life and pain were analysed in all randomised patients according to assigned treatment (excluding crossover treatment). Baseline was the last available pre-treatment assessment (note that questionnaires were not completed at screening, so this was typically at cycle 1). Time to worsening in FACT-P total score was defined as the time from randomisation to a decrease of 10 points or more,[1,2] clinical disease progression, or death. Time to worsening in BPI-SF pain intensity was defined as time from randomisation to an increase of 30% or more, an increase of 2 points or more,[3,4] clinical disease progression (excluding radiographic and PSA progression), or death. The stratified Cox regression model was used to estimate hazard ratio and

associated confidence interval (CI). Median, percentiles, and associated CI were estimated using the Kaplan–Meier method.

SUPPLEMENTARY RESULTS

Grade 5 treatment-related adverse events

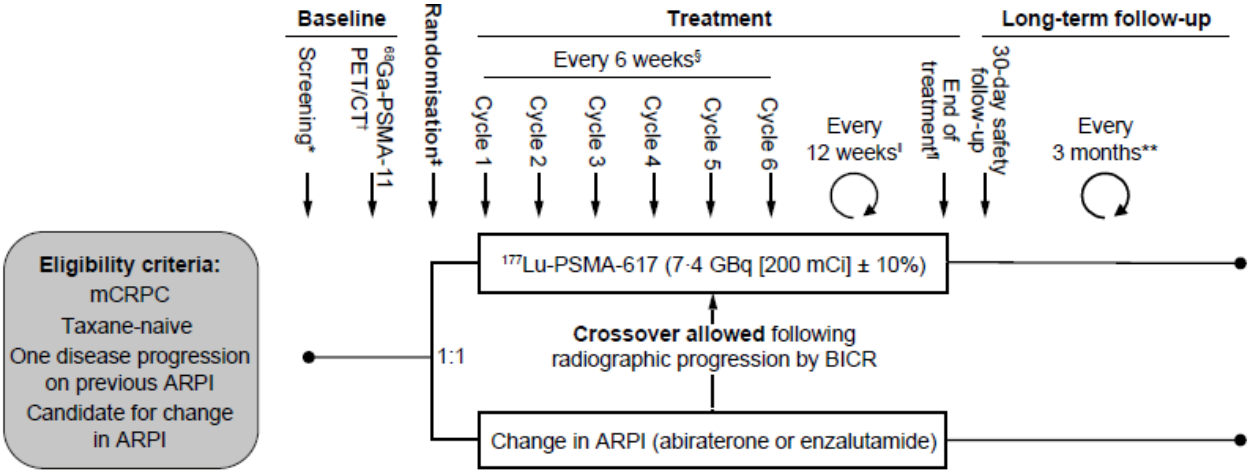
There were no grade 5 treatment-related adverse events in the ^{177}Lu-PSMA-617 group (0/227 patients) and one in the ARPI change group (1/232 patients; cerebrovascular accident).

^{68}Ga-PSMA-11 safety

^{68}Ga-PSMA-11 treatment-emergent adverse events are shown in Table S9.

SUPPLEMENTARY FIGURES

Figure S1: **Trial design**



CT/MRI and ^{99m}Tc bone scans	X††		... Every 8 weeks for 24 weeks, then every 12 weeks ...						X			X
ECOG performance status	X		X							X	X	
HRQoL and pain scores‡‡	X§§		X	X	X	X	X	X	X	X	X	X‖‖
Physical examination¶¶	X		X	X	X	X	X	X	X	X	X	X
Weight	X									X	X	
Vital signs	X		X	X	X	X	X	X	X	X	X	
Testosterone and PSA	X			X	X	X	X	X	X	X	X	X***
Haematology and biochemistry	X		X	X	X	X	X	X	X	X	X	X
Adverse events	X	X	X	X	X	X	X	X	X	X	X	X†††

* 14–42 days before starting treatment.

† Centrally read according to criteria defined in the trial protocol.

‡ Within 14 days before start of treatment. Stratification by previous ARPI use setting (castration-resistant *vs* hormone-sensitive) and pain symptoms (score 0–3 *vs* >3 on item 3 of the BPI-SF).

§ Assessments every week in cycle 1 (±3 days) and every 2 weeks in cycles 2–6 (±7 days).

‖ For patients assigned to ARPI change only. ARPI is administered continuously. Visit frequency is every 6 weeks for first 7 visits and every 12 weeks thereafter.

¶ Randomly assigned study treatment continued until completion of six cycles of ^{177}Lu-PSMA-617, radiographic disease progression, unacceptable toxicity, requirement for a prohibited treatment, non-compliance, lack of clinical benefit, or withdrawal by the patient, investigator, or sponsor.

** Follow-up until death, loss to follow-up, withdrawal of consent (including opposition to use of data or biological samples), or accrual of the number of events required for planned final analysis of overall survival.

†† Within 28 days of randomisation.

‡‡ EQ-5D-5L, FACT-P, and BPI-SF.

§§ BPI-SF only.

‖‖ 168 days and 336 days after safety follow-up.

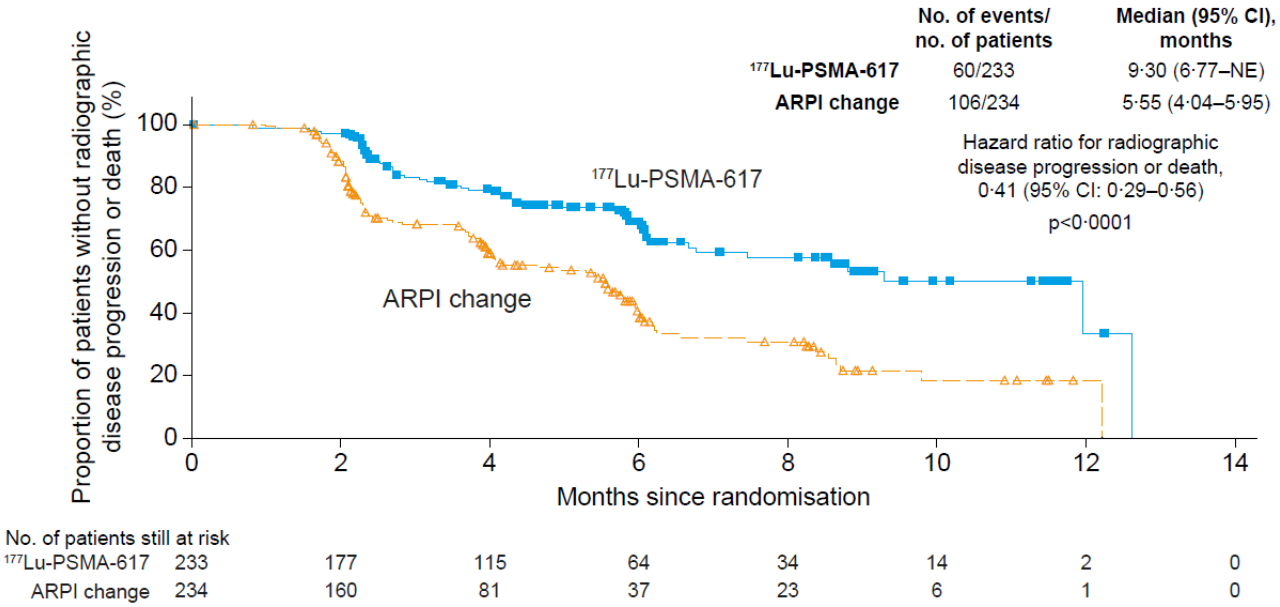
¶¶ Complete at baseline (general appearance, skin, neck [including thyroid], ears, eyes, nose, throat, lungs, heart, abdomen, back, lymph nodes, extremities, vascular, and neurological) and short thereafter (general appearance, blood pressure, and pulse).

*** PSA only.

††† For patients who received ^{177}Lu-PSMA-617 (including crossover), the incidences of haematologic toxicities (including transfusion or use of growth factors), renal failure, secondary malignancies, xerostomia, and xerophthalmia were assessed regardless of relationship to study treatment and whether a new anticancer therapy had been started.

ARPI=androgen receptor pathway inhibitor; BPI-SF=Brief Pain Inventory-Short Form; CT=computed tomography; ECOG=Eastern Co-operative Oncology Group; EQ-5D=EuroQol 5-Dimension 5-Level; FACT-P=Functional Assessment of Cancer Therapy-Prostate; HRQoL=health-related quality of life; mCRPC=metastatic castration-resistant prostate cancer; MRI=magnetic resonance imaging; PET=positron emission tomography; PSA=prostate-specific antigen; PSMA=prostate-specific membrane antigen.

Figure S2: **Primary analysis of radiographic progression-free survival at the time of first data cutoff (primary endpoint)**

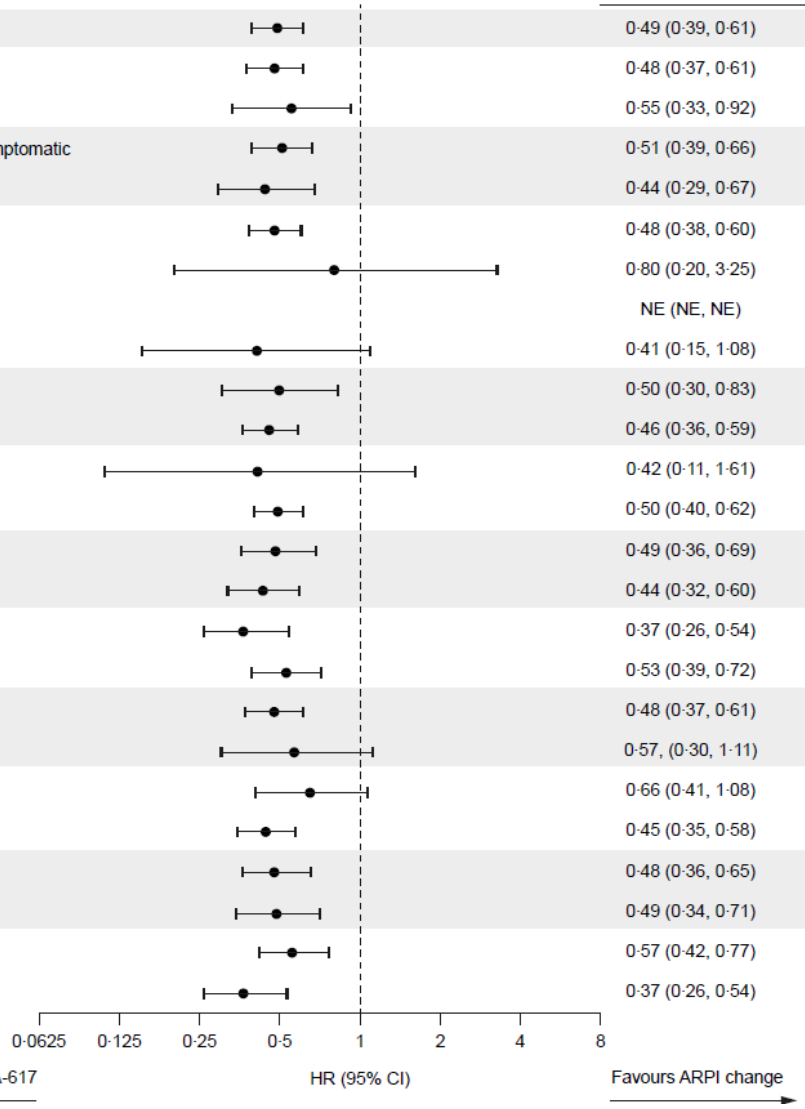


No. of patients still at risk								
^{177}Lu-PSMA-617	233	177	115	64	34	14	2	0
ARPI change	234	160	81	37	23	6	1	0

Data are from the primary analysis of the primary endpoint. At data-cut off, one patient had not yet been randomised, and was subsequently randomised to the ^{177}Lu-PSMA-617 group.

ARPI=androgen receptor pathway inhibitor; CI=confidence interval; NE=not estimable; PSMA=prostate-specific membrane antigen.

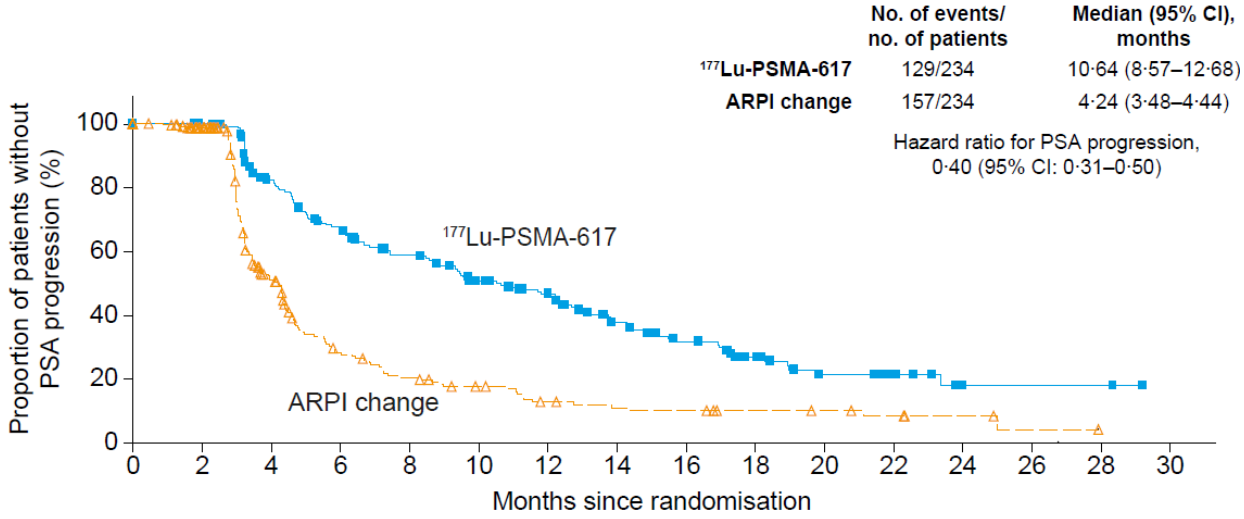
Figure S3: **Prespecified subgroup analysis of radiographic progression-free survival at the time of third data cutoff**



		HR (95% CI)	Events, n/n (%)	
			^{177}Lu-PSMA-617	ARPI change
All patients		0·49 (0·39, 0·61)	154/234 (66)	180/234 (77)
Previous ARPI setting	Castration-resistant	0·48 (0·37, 0·61)	125/190 (66)	147/190 (77)
	Hormone-sensitive	0·55 (0·33, 0·92)	29/44 (66)	33/44 (75)
Symptomatology	Asymptomatic or mildly symptomatic	0·51 (0·39, 0·66)	107/165 (65)	130/166 (78)
	Symptomatic	0·44 (0·29, 0·67)	47/69 (68)	50/68 (74)
OMB category	White	0·48 (0·38, 0·60)	137/212 (65)	164/214 (77)
	Black or African American	0·80 (0·20, 3·25)	5/7 (71)	4/5 (80)
	Asian	NE (NE, NE)	0/1 (0)	1/2 (50)
	Other	0·41 (0·15, 1·08)	12/14 (86)	11/13 (85)
Age	<65 years	0·50 (0·30, 0·83)	36/50 (72)	33/44 (75)
	≥65 years	0·46 (0·36, 0·59)	118/184 (64)	147/190 (77)
Liver metastases at baseline	Yes	0·42 (0·11, 1·61)	5/7 (71)	9/10 (90)
	No	0·50 (0·40, 0·62)	149/227 (66)	171/224 (76)
Baseline prostate-specific antigen level	<median	0·49 (0·36, 0·69)	62/106 (58)	89/122 (73)
	≥median	0·44 (0·32, 0·60)	88/123 (72)	86/106 (81)
Initial Gleason score	<8	0·37 (0·26, 0·54)	52/89 (58)	86/112 (77)
	≥8	0·53 (0·39, 0·72)	97/137 (71)	86/110 (78)
Baseline lactate dehydrogenase level	≤260 IU/L	0·48 (0·37, 0·61)	116/177 (66)	149/195 (76)
	>260 IU/L	0·57, (0·30, 1·11)	26/37 (70)	17/22 (77)
Region	North America	0·66 (0·41, 1·08)	32/47 (68)	38/52 (73)
	Europe	0·45 (0·35, 0·58)	122/187 (65)	142/182 (78)
ECOG performance status	0	0·48 (0·36, 0·65)	96/146 (66)	94/116 (81)
	1	0·49 (0·34, 0·71)	56/86 (65)	84/114 (74)
Previous ARPI	Abiraterone	0·57 (0·42, 0·77)	79/119 (66)	94/130 (72)
	Enzalutamide	0·37 (0·26, 0·54)	59/94 (63)	69/83 (83)

ARPI=androgen receptor pathway inhibitor; CI=confidence interval; HR=hazard ratio; NE=not estimable; OMB=Office of Management and Budget; PSMA=prostate-specific membrane antigen.

Figure S4: **Time to PSA progression (exploratory outcome) at the time of third data cutoff**



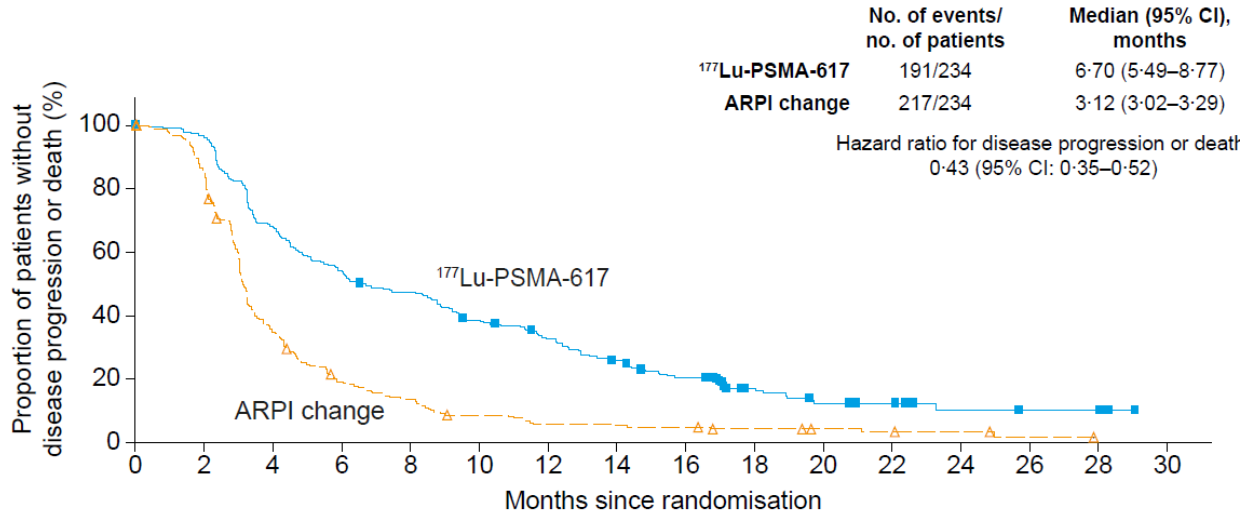
No. of patients still at risk

^{177}Lu-PSMA-617	234	211	160	126	103	79	67	46	35	23	13	9	2	2	2	0
ARPI change	234	206	89	44	31	23	15	12	11	8	7	5	3	1	0	0

PSA progression was defined as the time from randomisation to the following: ≥25% increase and ≥2 ng/mL above the nadir, confirmed by second value ≥3 weeks later if there was decline from baseline; or ≥25% increase and ≥2 ng/mL increase from baseline beyond 12 weeks if there was no decline from baseline. Assessments were at 3, 6, and 12 months after randomisation.

ARPI=androgen receptor pathway inhibitor; CI=confidence interval; PSA=prostate-specific antigen; PSMA=prostate-specific membrane antigen.

Figure S5: **Progression-free survival (secondary outcome) at the time of third data cutoff**

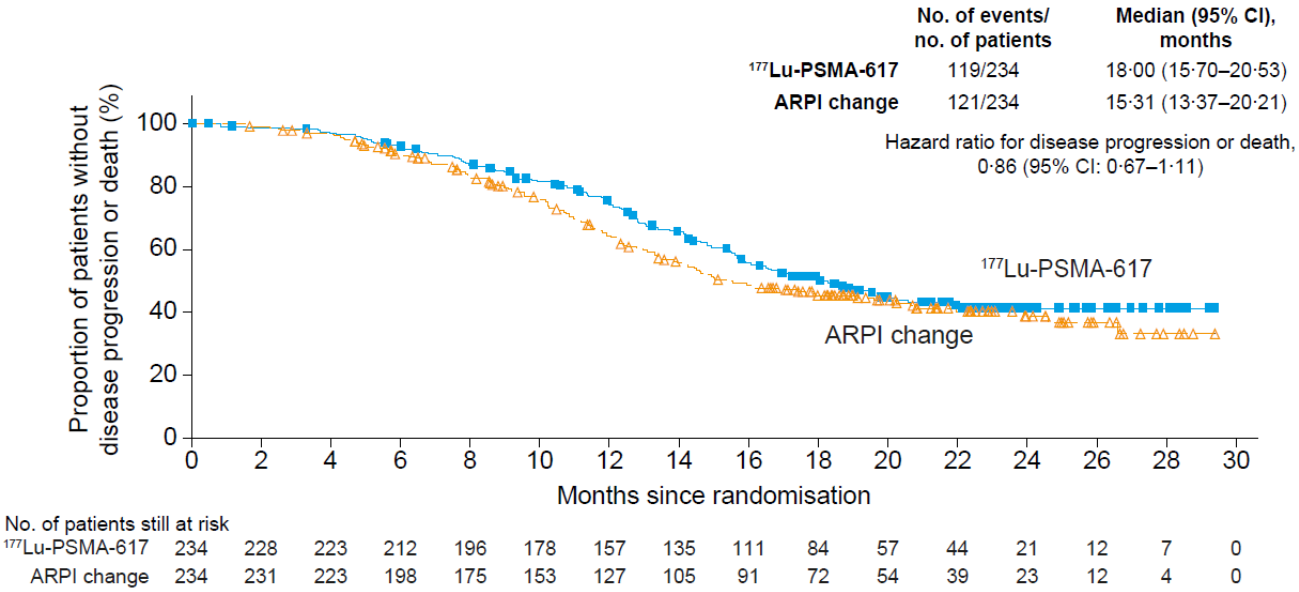


No. of patients still at risk

^{177}Lu-PSMA-617	234	220	155	123	107	86	71	55	42	22	15	13	5	4	4	0
ARPI change	234	196	79	42	30	18	12	11	10	7	5	4	3	1	0	0

ARPI=androgen receptor pathway inhibitor; CI=confidence interval; PSMA=prostate-specific membrane antigen.

Figure S6: **Progression-free survival on next line of treatment (secondary outcome) at the time of third data cutoff**



No. of patients still at risk

^{177}Lu-PSMA-617	234	228	223	212	196	178	157	135	111	84	57	44	21	12	7	0
ARPI change	234	231	223	198	175	153	127	105	91	72	54	39	23	12	4	0

ARPI=androgen receptor pathway inhibitor; CI=confidence interval; PSMA=prostate-specific membrane antigen.

SUPPLEMENTARY TABLES

Table S1: Prespecified endpoints and definitions

Endpoint	Definition
Primary endpoint	
Radiographic progression-free survival	Time from randomisation to first documented radiographic disease progression according to PCWG3-modified RECIST v1·1 assessed by BICR, or death
Key secondary endpoint	
Overall survival	Time from randomisation to death
Additional secondary endpoints	
Second radiographic progression-free survival	Time from crossover to radiographic disease progression (as defined above) or death
Progression-free survival	Time from randomisation to disease progression by investigator assessment (radiographic, clinical, or PSA) or death
Second progression-free survival	Time from randomisation to disease progression on next line of therapy by investigator assessment (radiographic, clinical, or PSA) or death
PSA response	Proportion of patients with a ≥50% PSA decrease from baseline, evaluated at months 3, 6, and 12, and confirmed by a second assessment within 4 weeks
Time to symptomatic skeletal event	Time from randomisation to first new symptomatic pathological bone fracture, spinal cord compression, tumour-related orthopaedic surgery, radiation therapy for bone pain, or death
Time to soft tissue progression	Time from randomisation to radiographic soft tissue progression according to PCWG3-modified RECIST v1·1 assessed by BICR
Time to chemotherapy	Time from randomisation to initiation of the first subsequent chemotherapy or death
Health-related quality of life	Assessed using patient-reported EQ-5D-5L, FACT-P, and BPI-SF (see Supplementary Methods)
Safety and tolerability	Frequency of adverse events, safety assessments, and vital signs
Exploratory endpoints	
Objective response rate	Proportion of patients with complete or partial response as per PCWG3-modified RECIST v1·1 assessed by BICR
Disease control rate	Proportion of patients with complete or partial response or stable disease as per PCWG3-modified RECIST v1·1 assessed by BICR
Duration of response	Length of time between first documented best overall response as per PCWG3-modified RECIST v1·1 assessed by BICR, and the first documented disease progression or death
Time to PSA progression	Time from randomisation to PSA progression (defined as ≥25% increase and ≥2 ng/mL above the nadir, confirmed by second value ≥3 weeks later if there is decline from baseline, or ≥25% increase and ≥2 ng/mL increase from baseline beyond 12 weeks if there is no decline from baseline)
Time to pain progression	Time from randomisation to pain progression as assessed by patient-reported BPI-SF
Biomarkers	Molecular analysis of tumour tissue and liquid biopsy through targeted or multiplex assays and association with clinical efficacy endpoints

BICR=blinded independent central review; BPI-SF=Brief Pain Inventory-Short Form; EQ-5D-5L=EuroQoL 5-Domain 5-Level; FACT-P=Functional Assessment of Cancer Therapy-Prostate; PCWG3=Prostate Cancer Working Group 3; PSA=prostate-specific antigen; RECIST=Response Evaluation Criteria in Solid Tumours.

Table S2: **Protocol deviations at the time of third data cutoff**

Protocol deviation	^{177}Lu-PSMA-617 (n=234)	ARPI change (n=234)
Any – no. (%)	202 (86)	185 (79)
Other GCP deviation – no. (%)	171 (73)	168 (72)
Treatment deviation – no. (%)[*,†]	70 (30)	7 (3)
Participant not withdrawn as per protocol – no. (%)[*,‡]	47 (20)	19 (8)
Any exclusion criteria deviation – no. (%)	14 (6)	13 (6)
Any inclusion criteria deviation – no. (%)	7 (3)	16 (7)
Prohibited concomitant medication – no. (%)	2 (1)	1 (0)

Each protocol deviation category consists of multiple deviation terms.

A patient with multiple protocol deviations is counted only once per category.

Patients may have protocol deviations in more than one category.

Protocol deviations following crossover from ARPI change to ^{177}Lu-PSMA-617 are not included.

For re-screened patients, any deviations recorded during initial screening are not included.

* Category disproportionately affected by ^{177}Lu-PSMA-617 production halt.

† Treatment deviation includes: >110% or <90% of target ^{177}Lu-PSMA-617 dose received; ^{177}Lu-PSMA-617 or ARPI not received on Cycle 1 Day 1 within 14+3 day window of randomization or registration for crossover; non-compliant preparation and administration of ^{68}Ga or ^{177}Lu, exclusive of dose given; treatment not modified for toxicity per protocol; treatment not given because of COVID-19.

‡ Participant not withdrawn as per protocol includes: continuation despite ^{177}Lu-PSMA-617 or ARPI interruption of longer than 4 weeks; continuation on study treatment despite no clinical benefit after evidence of tumour progression as measured by PCWG3-modified RECIST V1.1 criteria; continuation of sample and data collection after participant consent withdrawn.

ARPI=androgen receptor pathway inhibitor; COVID-19=coronavirus disease 2019; GCP=Good Clinical Practice; PCWG3=Prostate Cancer Working Group 3; PSMA=prostate-specific membrane antigen; RECIST=Response Evaluation Criteria in Solid Tumours.

Table S3: **Post-protocol anticancer therapies at the time of third data cutoff**

	^{177}Lu-PSMA-617 (n=234)	Post-crossover[*] (n=134)	ARPI change[†] (n=234)
Radiotherapy – no. (%)	44 (19)	20 (15)	41 (18)
Anticancer medication (excluding crossover)			
Chemotherapy	110 (47)	58 (43)	99 (42)
Hormonal therapy	31 (13)	4 (3)	5 (2)
Biologic therapy	8 (3)	0	3 (1)
Targeted therapy (excluding radioligand therapy)[‡]	8 (3)	9 (7)	11 (5)
Radium-223 dichloride	1 (0)	2 (1)	4 (2)
Investigational anticancer drugs	2 (1)	0	0
^{177}Lu-PSMA-617	1 (0)	0	8 (3)
Other radiopharmaceuticals	3 (1)	1 (1)	1 (0)
Other	5 (2)	0	0

* Post-crossover therapy does not include crossover ^{177}Lu-PSMA-617.

† Includes all patients randomised to ARPI change including those who crossed over.

‡ Capivasertib (n=3 in the ^{177}Lu-PSMA-617 group; n=4 in the post-crossover group), olaparib (n=4 in the ^{177}Lu-PSMA-617 group; n=3 in the post-crossover group; n=4 in the ARPI change group), abemaciclib (n=1 in the post-crossover group), niraparib (n=1 in the ARPI change group), or investigational drug (n=1 in the ^{177}Lu-PSMA-617 group; n=1 in the post-crossover group).

ARPI=androgen receptor pathway inhibitor; PSMA=prostate-specific membrane antigen.

Table S4: **Exposure to study drugs***

	^{177}Lu-PSMA-617 (n=227)	Abiraterone (n=100)	Enzalutamide (n=132)	ARPI change (n=232)
Median duration of exposure (IQR) – months				
At time of primary analysis	4·25 (2·40–7·10)†	4·20 (2·10–6·30)	3·55 (1·90–6·60)	3·90 (2·00–6·35)
At time of third interim overall survival analysis	8·41 (6·21–9·17)	6·36 (3.88–10.32)	6·59 (3·12–12·14)	6·52 (3·29–11·06)
^{177}Lu-PSMA-617 injections				
Median number of injections at time of primary analysis (IQR)	3·0 (2·0–5·0)	NA	NA	NA
At least 1 injection – no. (%)	226 (100)†	NA	NA	NA
At least 2 injections – no. (%)	195 (86)†	NA	NA	NA
At least 3 injections – no. (%)	141 (62)†	NA	NA	NA
At least 4 injections – no. (%)	107 (47)†	NA	NA	NA
At least 5 injections – no. (%)	69 (31)†	NA	NA	NA
6 injections – no. (%)	40 (18)†	NA	NA	NA
Median number of injections at time of third interim overall survival analysis (IQR)	6·0 (4·0–6·0)	NA	NA	NA
At least 1 injection – no. (%)	227 (100)	NA	NA	NA
At least 2 injections – no. (%)	220 (97)	NA	NA	NA
At least 3 injections – no. (%)	197 (87)	NA	NA	NA
At least 4 injections – no. (%)	180 (79)	NA	NA	NA
At least 5 injections – no. (%)	165 (73)	NA	NA	NA
6 injections – no. (%)	144 (63)	NA	NA	NA
At least one dose decrease or interruption – no. (%)				
At time of primary analysis	61 (27)†	10 (10)	35 (27)	45 (19)
At time of third interim overall survival analysis	85 (37)	14 (14)	46 (35)	60 (26)
At least one dose interruption, – no. (%)				
At time of primary analysis	48 (21)†	10 (10)	19 (14)	29 (13)
At time of third interim overall survival analysis	62 (27)	14 (14)	25 (19)	39 (17)
At least one dose decrease – no. (%)				
At time of primary analysis	18 (8)†	4 (4)	30 (23)	34 (15)
At time of third interim overall survival analysis	29 (13)	4 (4)	39 (30)	43 (19)
Permanent discontinuation – no. (%)				
At time of primary analysis	85 (38)†	50 (5)	60 (45)	110 (47)
At time of third interim overall survival analysis	227 (100)	96 (96)	116 (88)	212 (91)
Median dose intensity (IQR)‡				
At time of primary analysis	5·20 (4·80–5·70)†	1000·00 (1000·00–1000·00)	160·00 (157·00–160·00)	NA
At time of third interim overall survival analysis	4·95 (4·61–5·19)	1000·00 (1000·00–1000·00)	160·00 (150·83–160·00)	NA

* Data from patients who underwent randomisation and received at least one dose of their assigned treatment.

† n=226.

‡ Units are GBq/month for ^{177}Lu-PSMA-617 and mg/day for abiraterone and enzalutamide.

ARPI=androgen receptor pathway inhibitor; IQR=interquartile range; NA=not applicable; PSMA=prostate-specific membrane antigen.

Table S5: **Time to symptomatic skeletal events at the time of third data cutoff (secondary outcome)**

	^{177}Lu-PSMA-617 (n=234)	**ARPI change (n=234)**
Events – no. (%)	31 (13)	68 (29)
Symptomatic skeletal event	27 (12)	63 (27)
Symptomatic pathological bone fracture	4 (2)	13 (6)
Spinal cord compression	6 (3)	4 (2)
Tumour-related orthopaedic surgical intervention	1 (0)	3 (1)
Requirement for radiation therapy to relieve bone pain	16 (7)	43 (18)
Death – no. (%)*	4 (2)	5 (2)
Censored – no. (%)	203 (87)	166 (71)
Median (95% CI) – months	NE (NE–NE)	17·97 (14·26–NE)
HR (95% CI)†	0·41 (0·26–0·63)	

* Death without symptomatic skeletal event.

† Stratified Cox proportional-hazards model.

ARPI=androgen receptor pathway inhibitor; CI=confidence interval; HR=hazard ratio; NE=not estimable; PSMA=prostate-specific membrane antigen.

Table S6: **RECIST v1·1 responses as per blinded independent central review at the time of third data cutoff (exploratory outcome)**

	^{177}Lu-PSMA-617	ARPI change
Patients with measurable soft tissue disease at baseline, excluding bone lesion progression		
Number of patients	72	72
Best overall response – no (%)		
Complete soft tissue response	15 (21)	3 (4)
Partial soft tissue response	21 (29)	8 (11)
Stable soft tissue disease	23 (32)	32 (44)
Progressive soft tissue disease	11 (15)	24 (33)
Unknown	2 (3)	5 (7)
Objective response – no. (%)*	36 (50)	11 (15)
95% CI	38·0–62·0	7·9–25·7
Disease control – no. (%)[†]	59 (82)	43 (60)
95% CI	71·1–90·0	47·5–71·1
Median duration of response (95% CI) – months[‡§]	NE (15·67–NE)	10·05 (4·63–NE)
Patients with measurable soft tissue disease at baseline, including bone lesion progression		
Number of patients	72	72
Best overall response – no (%)		
Complete response	15 (21)	2 (3)
Partial response	20 (28)	8 (11)
Stable disease	21 (29)	30 (42)
Progressive disease	14 (19)	27 (38)
Unknown	2 (3)	5 (7)
Objective response – no. (%)*	35 (49)	10 (14)
95% CI	36·7–60·7	6·9–24·1
Disease control – no. (%)[†]	56 (78)	40 (56)
95% CI	66·4–86·7	43·4–67·3
Patients with evaluable soft tissue disease at baseline, excluding bone lesion progression		
Number of patients	104	102
Best overall response – no (%)		
Complete soft tissue response	22 (21)	6 (6)
Partial soft tissue response	21 (20)	8 (8)
Non-complete soft tissue response/non-progressive soft tissue disease	21 (20)	19 (19)
Stable soft tissue disease	25 (24)	32 (31)
Progressive soft tissue disease	13 (13)	30 (29)
Unknown	2 (2)	7 (7)
Objective response – no. (%)*	43 (41)	14 (14)
95% CI	31·8–51·4	7·7–22·0
Disease control – no. (%)[†]	89 (86)	65 (64)
95% CI	77·3–91·7	53·6–73·0
Median duration of response (95% CI) – months[‡‖]	NE (17·12–NE)	13·90 (4·63–NE)
Patients with evaluable soft tissue disease at baseline, including bone lesion progression		
Number of patients	104	102
Best overall response – no (%)		
Complete response	22 (21)	4 (4)

Partial response	20 (19)	8 (8)
Non-complete response/non-progressive disease	20 (19)	17 (17)
Stable disease	23 (22)	30 (29)
Progressive disease	17 (16)	37 (36)
Unknown	2 (2)	6 (6)
Objective response – no. (%)*	42 (40)	12 (12)
95% CI	30·9–50·5	6·2–19·6
Disease control – no. (%)†	85 (82)	59 (58)
95% CI	72·9–88·6	47·7–67·6

* Sum of complete response and partial response.

† Sum of complete response, partial response, and stable disease (plus non-complete response/non-progressive disease for patients without measurable disease).

‡ Estimated using the Kaplan–Meier method in patients with objective response as the duration from first documented best overall response to documented disease progression or death.

§ ^{177}Lu-PSMA-617, n=36; ARPI change, n=11.

ǀ ^{177}Lu-PSMA-617, n=43; ARPI change, n=14.

ARPI=androgen receptor pathway inhibitor; CI=confidence interval; NE=not estimable; PSMA=prostate-specific membrane antigen; RECIST v1·1=Response Evaluation Criteria in Solid Tumours version 1·1.

Table S7: **Treatment-emergent adverse events (expanded)***

Treatment-emergent adverse event	^{177}Lu-PSMA-617 (n=227)		ARPI change (n=232)	
	Any grade	Grade ≥3	Any grade	Grade ≥3
Any – no. (%)	224 (99)	81 (36)[†]	226 (97)	112 (48)[†]
Any drug-related – no. (%)	199 (88)	27 (12)	148 (64)	29 (13)
Serious – no. (%)	46 (20)	41 (18)	75 (32)	69 (30)
Serious drug-related – no. (%)	6 (3)	5 (2)	6 (3)	6 (3)
Leading to death – no. (%)[‡]	4 (2)	4 (2)	5 (2)	5 (2)
Leading to death, drug-related – no. (%)	0	0	1 (0)	1 (0)
Leading to adjustment of dose – no. (%)	8 (4)	3 (1)	36 (16)	7 (3)
Leading to interruption of treatment – no. (%)	28 (12)	13 (6)	45 (19)	20 (9)
Leading to discontinuation of treatment – no. (%)	13 (6)	8 (4)	12 (5)	9 (4)
Occurring in ≥5% of patients in the ^{177}Lu-PSMA-617 group – no. (%)				
Dry mouth	131 (58)[§]	2 (1)	6 (3)[§]	0
Asthenia	74 (33)	2 (1)	67 (29)	8 (3)
Nausea	72 (32)	0	27 (12)	1 (0)
Anaemia	61 (27)	14 (6)	44 (19)	16 (7)
Fatigue	53 (23)	1 (0)	59 (25)	4 (2)
Constipation	50 (22)	1 (0)	33 (14)	0
Decreased appetite	49 (22)	0	43 (19)	1 (0)
Arthralgia	45 (20)	0	54 (23)	1 (0)
Diarrhoea	38 (17)	0	21 (9)	1 (0)
COVID-19	36 (16)	1 (0)	27 (12)	1 (0)
Back pain	31 (14)	3 (1)	46 (20)	6 (3)
Vomiting	26 (11)	0	11 (5)	0
Oedema peripheral	19 (8)	0	28 (12)	0
Dysgeusia	19 (8)	0	6 (3)	0
Headache	19 (8)	0	11 (5)	0
Pain in extremity	17 (7)	1 (0)	25 (11)	1 (0)
Thrombocytopenia	17 (7)	5 (2)	7 (3)	2 (1)
Bone pain	16 (7)	1 (0)	16 (7)	2 (1)
Weight loss	15 (7)	1 (0)	32 (14)	3 (1)
Blood alkaline phosphatase increased	15 (7)	8 (4)	17 (7)	6 (3)
Cough	15 (7)	0	11 (5)	0
Hot flush	14 (6)	0	21 (9)	0
Dry eye	13 (6)	0	2 (1)	0
Myalgia	13 (6)	0	12 (5)	1 (0)
Neutropenia	13 (6)	3 (1)	2 (1)	1 (0)
Urinary tract infection	12 (5)	3 (1)	12 (5)	5 (2)
Abdominal pain	12 (5)	2 (1)	11 (5)	1 (0)

* Data are for patients who underwent randomisation and received at least one dose of their assigned treatment; 7 patients who were randomly allocated to the ^{177}Lu-PSMA-617 group did not receive treatment. Treatment-emergent adverse events were those occurring upon or after the start of randomised treatment and up to 30 days after the last dose of ARPI or up to 41 days after the last dose of ^{177}Lu-PSMA-617. Adverse events were coded using Common Terminology Criteria for Adverse Events v5·0 grading and Medical Dictionary for Regulatory Activities v26·0 terms.

[†] Grade 3–4: n=80 (35%) in the ^{177}Lu-PSMA-617 group; n=111 (48%) in the ARPI change group.

[‡] In the ^{177}Lu-PSMA-617 group: COVID-19 pneumonia, cardiac arrest, intestinal ischemia, and sepsis (all n=1; all considered not related to study drug by the investigator); in the ARPI change group: cardiac arrest, coma, dyspnoea, multiple organ dysfunction syndrome (all n=1; all considered not related to study drug by the investigator), and cerebrovascular accident (n=1; considered related to study drug by the investigator).

[§] Grade 2 dry mouth: n=19 in ^{177}Lu-PSMA-617 group; n=0 in ARPI change group.

ARPI=androgen receptor pathway inhibitor; PSMA=prostate-specific membrane antigen.

Table S8: Treatment-emergent safety topics of interest

Safety topic (occurring in more than one patient)	^{177}Lu-PSMA-617 (n=227)		ARPI change (n=232)	
	Any grade	Grade ≥3	Any grade	Grade ≥3
Safety topic – no. (%)				
Dry mouth*	138 (61)	2 (1)	6 (3)	0
Myelosuppression[†]	86 (38)	30 (13)	54 (23)	20 (9)
Hepatoxicity[‡]	30 (13)	13 (6)	31 (13)	11 (5)
Renal toxicity[§]	15 (7)	3 (1)	18 (8)	7 (3)
Dry eye[l]	14 (6)	0	2 (1)	0
Fractures[¶]	9 (4)	2 (1)	18 (8)	11 (5)
QT prolongation**	4 (2)	1 (0)	6 (3)	4 (2)
Second primary malignancies[††]	4 (2)	2 (1)	3 (1)	2 (1)
Intracranial haemorrhage[‡‡]	3 (1)	1 (0)	1 (0)	1 (0)

Safety topics of interest are groupings of treatment-emergent adverse events (by Medical Dictionary for Regulatory Events v26·0 terms).

Treatment-emergent adverse events were those occurring upon or after the start of randomised treatment and up to 30 days after the last dose of ARPI or up to 41 days after the last dose of ^{177}Lu-PSMA-617.

Adverse events were coded using Common Terminology Criteria for Adverse Events v5·0 grading and Medical Dictionary for Regulatory Activities v26·0 terms.

Data are for patients who underwent randomisation and received at least one dose of their assigned treatment; 7 patients who were randomly allocated to the ^{177}Lu-PSMA-617 group did not receive treatment.

* Dry mouth includes: dry mouth, aptyalism, dry throat, lip dry.

[†] Myelosuppression includes: anaemia, thrombocytopenia, neutropenia, white blood cell count decreased, lymphocyte count decreased, platelet count decreased, leukopenia, lymphopenia, neutrophil count decreased, febrile neutropenia, haemoglobin decreased, pancytopenia.

[‡] Hepatoxicity includes: blood alkaline phosphatase increased, aspartate aminotransferase increased, alanine aminotransferase increased, γ-glutamyl transferase increased, hypertransaminasemia, blood bilirubin increased, hepatic encephalopathy, hepatic enzyme increased, hypoalbuminemia, international normalized ratio increased, ascites, hepatic cytolysis, transaminases increased.

[§] Renal toxicity includes: blood creatinine increased, acute kidney injury, renal failure (n=2 in the ^{177}Lu-PSMA-617 group; n=3 in the ARPI change group), blood urea increased.

[l] Dry eye includes: dry eye, xerophthalmia.

[¶] Fractures includes: femur fracture, rib fracture, ankle fracture, costal cartilage fracture, foot fracture, lumbar vertebral fracture, pathological fracture, shoulder fracture, acetabulum fracture, cervical vertebral fracture, radius fracture, thoracic vertebral fracture, wrist fracture.

** QT prolongation includes: cardiac arrest, electrocardiogram QT prolonged, loss of consciousness, syncope.

[††] Second primary malignancies includes: adenocarcinoma of colon, basal cell carcinoma, clear cell renal carcinoma, renal cell carcinoma, leiomyosarcoma, leiomyosarcoma metastatic, transitional cell carcinoma.

[‡‡] Intracranial haemorrhage includes: cerebrovascular accident, haemorrhage intracranial.

ARPI=androgen receptor pathway inhibitor; PSMA=prostate-specific membrane antigen.

Table S9: **Treatment-emergent adverse events for ^{68}Ga-PSMA-11***

Treatment-emergent adverse event	Patients who received ^{68}Ga-PSMA-11 (n=547)	
	Any grade	Grade ≥3
Any – no. (%)	65 (12)	8 (1)
Any drug-related – no. (%)	9 (2)	0
Serious – no. (%)	8 (1)	8 (1)
Serious drug-related – no. (%)	0	0
Leading to death – no. (%)	0	0
Occurring in ≥0.5% patients – no. (%)		
Back pain	6 (1)	0
Anaemia	6 (1)	0
Arthralgia	5 (1)	0
Diarrhoea	4 (1)	1 (0)
Decreased appetite	4 (1)	0
Asthenia	3 (1)	1 (0)
Constipation	3 (1)	0
Hot flush	3 (1)	0

Treatment-emergent adverse events were those occurring upon or after date of ^{68}Ga-PSMA-11 dosing and up to 14 days afterwards provided this is prior to first dose of ^{177}Lu-PSMA-617.

Adverse events were coded using Common Terminology Criteria for Adverse Events v5·0 grading and Medical Dictionary for Regulatory Activities v26·0 terms.

* Data are for all patients who received a dose of ^{68}Ga-PSMA-11, including those not enrolled.

PSMA=prostate-specific membrane antigen.

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